

Chance-Constrained Frequency Dynamic Unit Commitment Considering Uncertain Variability through Bayesian Filter Method

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Abstract—Ensuring frequency dynamic security with considering uncertain variability (UV) is a critical challenge in low-inertia grids with a high penetration of renewable energy sources. In this situation, a Bayesian-filer-derived system frequency response (BF-SFR) method and a well-designed chance constrained frequency dynamic unit commitment (CC-FDUC) model are proposed to solve this problem. A Bayesian filter method for modeling SFR is created after the UV induced by numerous converter-interfaced resources and potential measured errors is examined. Unlike previous approaches such as analytical SFR method and time-discretized ones without considering UV, BF-SFR offers the trajectory of frequency derivation as existing methods do, it additionally estimate the posterior distribution of frequency derivation if UV incorporated. Following this, The posterior distribution through BF-SFR is reformulated as a set of chance constrained frequency dynamic restrictions in CC-FDUC to search optimal scheduled commitment results. Several cases on a modified IEEE 6 bus test system and a 118 bus test system confirm the unbiased and validness of the proposed models.

Index Terms—Bayesian filter, system frequency response, uncertain variability, unit commitment, chance constraints, converter-interfaced generators.

I. INTRODUCTION

THE UV characteristic from multiple converter-interfaced generators and measured errors has a negative impact on the frequency dynamic security and energy management in low-inertia grids. To keep the system have an adequate capacity against UV, many studies and state Grid Codes are reported to explore the necessity of grid-connected converter-interfaced generators participating into frequency supporting in order to preserve the intended system's acceptable capacity against UV.

The literature reported several studies to model SFR against imbalanced power disturbances. These studies used an center-of-inertia (CoI) swing equation to calculate frequency dynamics. Ref. [1, 2] built a low-order CoI equation concerning conventional synchronous generators and derived an analytical solution. Ref. [3-6] considered the impact of converter-interfaced generators with various frequency control schemes. It's worth noting that such models in [1-6] kept similar mathematical types of ordinary differential equation. As such, their solutions

kept similar forms, the difference of associated solutions lies in a modification of frequency dynamic parameters due to integrated converter-interfaced generators. The shortage of such solutions mentioned before lies in a lack of consideration of complex frequency control schemes. Driven by this issue, another type of time-discretized solution encoding SFR is developed. Ref. [7, 8] constructed a time-discretized solution considering rate-of-change-of-frequency (RoCoF) replays. Ref. [9] extended a multi-area time-discretized solution considering frequency supporting of fly-wheel storage systems. Ref. [11] derived a time-discretized SFR solution considering converter-based generators. The numerical result of frequency response, [7-11], reflects the trajectory of frequency dynamic with free noise assumptions. Nevertheless, the frequency dynamic with noise assumptions cannot be solved. Furthermore, the probabilistic distribution of the system frequency change cannot be well estimated.

In the application of frequency dynamic in electrical power system operations, some FDUC models were carried out on introducing frequency dynamic requirements derived from SFR into UC problems. These models can be divided into two categories in line with integrated frequency dynamic constraints. i) critical feature requirement of frequency dynamic, such as RoCoF, nadir, is reshaped as a set of convex constraints into UC [3, 5, 6, 11-17]; ii) frequency change over a predefined simulated horizon (e.g., 0s - 60s) is reformulated into a set of binding constraints incorporating into UC, [7, 9, 18-19]. Compared with the former category, the latter one is more accurate but more complicated. In despite that the involvement of conventional generators participating in UC has been thoroughly discussed, few studies focus frequency supporting of converter-interfaced generators and BESS from the probabilistic perspective.

Motivated by such issues, this paper firstly presents a new probabilistic Bayesian filter framework for encoding SFR with UV hypothesis. BF-SFR captures the interactive impact of UV on the frequency derivation, it can not only estimate the trajectory of frequency derivation but also provide its posterior probabilistic density distribution (PDF). Therefore, BF-SFR provides a great level of freedom for analyzing frequency dynamic and external factors such as UV, unit's output limits, multiple control interaction. Following this, a classical application, FDUC is implemented. In particular, the impact of frequency dynamic requirement with UV assumption within BF-SFR is extended, and chance-constrained FDUC model is well redesigned.

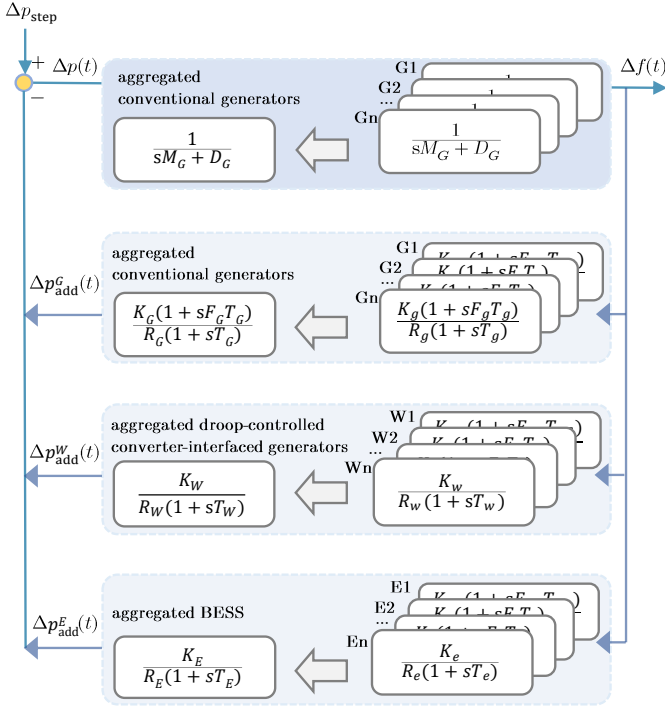


Figure 1. The unified frequency dynamic response of electrical power systems against occurred power disturbance

In all, the contributions in this paper can be summarized, i) A BF-SFR solution for modeling SFR is proposed. The impact of SFR considering UV is investigated and analyzed. The posterior probabilistic distribution of frequency derivation can be well estimated. Meanwhile, the effect of various control schemes provided by converter-interfaced generators and BESS can be reconsidered. ii) A CCFDUC model with considering frequency dynamic derived from BF-SFR is designed. A chance constraint form of frequency dynamic requirement concerning nadir is derived. The convex characteristic and linearization of constructed frequency dynamic constraint is discussed.

The reminder of this paper is organized: Section II introduces a Bayesian filter framework for encoding SFR and derives the solution of BF-SFR. Section III constructs a set of chance-constraint frequency dynamic constraints to enforce the impact of frequency dynamic requirements, and sequentially gives a CCFDUC model. Section IV verifies the effectiveness of BF-SFR and CCFDUC in a modified IEEE 14 bus test system and IEEE 118 bus test system, and Section V concludes on the main work in this paper.

II. MODELING BF-SFR THROUGH BAYESIAN-FILTER

Considering a low-inertia grid comprised of conventional generators, converter-interfaced generators and BESS, the union frequency dynamic of studied power system can be combined as Fig. 1(a) and simplified as Fig. 1(b). The impact of homogeneous generators in Fig. 1(a) with similar frequency control schemes can be equivalent as an aggregated CoI unit, illustrated in Fig. 1(b). As such, the aggregated coefficients $\{H_G, D_G, K_G, F_G, R_G, T_G\}$, $\{K_W, R_W, M_W, D_W, T_W\}$ and

$\{T_E, R_E\}$, respectively, represent adjusted output produced by CoI-based conventional unit block, CoI-based converter-interfaced block unit and CoI-based BESS block. More details on the derivation from individual unit's frequency parameters to aggregated frequency parameters can be found in [6].

In Fig. 1(b), the reduced order transfer function of SFR concerning a step-type mismatch power disturbance $\Delta p_{step}(s) = \Delta p_{step}/s$ can be rewritten as,

$$(M_G s + D_G) \Delta f(s) = (1/s \Delta p_{step}) - \underbrace{(K_G/R_G) [(1 + F_G T_G s)/(1 + T_G s)] \Delta f(s)}_{\text{CoI conventional unit}} - \underbrace{(1/R_W) (1/(1 + T_W s)) \Delta f(s)}_{\text{CoI Droop-controlled converter-interfaced unit}} - \underbrace{(M_E s + D_E) \Delta f(s)}_{\text{CoI BESS}} \quad (1)$$

The t -domain mathematical form of (1) can be transferred by utilizing the inverse Laplace transformation,

$$M_{\text{equ}} \Delta f'(t) + D_{\text{equ}} \Delta f(t) = \Delta p(t) \quad (2a)$$

$$\Delta p(t) = \Delta p_{\text{step}} - \Delta p_{\text{add}}^G(t) - \Delta p_{\text{add}}^W(t) - \Delta p_{\text{add}}^E(t) \quad (2b)$$

$$M_{\text{equ}} = M_G + M_W \quad (2c)$$

$$D_{\text{equ}} = D_G + D_W + R_W^{-1} + R_E^{-1} \quad (2d)$$

$$\Delta p_{\text{add}}^G(t) = K_G F_G R_G^{-1} \Delta f(t) - K_G (1 - F_G) R_G^{-1} e^{-\frac{1}{T_G} t} \int_0^t e^{\frac{1}{T_G} \kappa} \Delta f'(\kappa) d\kappa \quad (2e)$$

$$\Delta p_{\text{add}}^W(t) = (R_W^{-1}) \Delta f(t) \quad (2f)$$

$$\Delta p_{\text{add}}^E(t) = (R_E^{-1}) \Delta f(t) \quad (2g)$$

Detailed derivation on (1) $\rightarrow$ (2) can be found at appendix II. By replacing $\Delta f'(t)|_{t=t_0+n\Delta t}$ as its equivalent discretized form $[\Delta f(t_0 + n\Delta t) - \Delta f(t_0 + (n-1)\Delta t)]/\Delta t$, the time-discretized form of $\Delta f(t)$ can be well estimated as below,

$$\Delta f(t_0 + (n+1)\Delta t) = \Delta f(t_0 + n\Delta t) + \quad (3a)$$

$$M_{\text{equ}}^{-1} [\Delta p(t_0 + n\Delta t) - D_{\text{equ}} \Delta f(t_0 + n\Delta t)] \quad (3b)$$

$$\Delta p(t_0 + n\Delta t) = \Delta p_{\text{step}} - \Delta p_{\text{add}}^G(t_0 + n\Delta t) - \quad (3c)$$

$$\Delta p_{\text{add}}^W(t_0 + n\Delta t) - \Delta p_{\text{add}}^E(t_0 + n\Delta t)$$

$$\Delta p_{\text{add}}^G(t_0 + n\Delta t) = K_G R_G^{-1} \Delta f(t_0 + n\Delta t) - \quad (3d)$$

$$\left(K_G (1 - F_G) R_G^{-1} e^{-\frac{1}{T_G} (t_0 + n\Delta t)} \right) \times$$

$$\left(\sum_{i=0}^n e^{\frac{1}{T_G} (t_0 + i\Delta t)} [\Delta f(t_0 + i\Delta t) - \Delta f(t_0 + (i-1)\Delta t)] \right) \quad (3e)$$

$$\Delta p_{\text{add}}^W(t_0 + n\Delta t) = (R_W^{-1}) \Delta f(t_0 + n\Delta t) \quad (3e)$$

$$\Delta p_{\text{add}}^E(t_0 + n\Delta t) = (R_E^{-1}) \Delta f(t_0 + n\Delta t) \quad (3f)$$

where NT is a total number of simulated steps, Δt declares a predefined discretized step (in this paper, $\Delta t = 0.05s$). (3b) depicts the temporal evolution of frequency derivation between adjacent discretized steps; (3c) describes the derivation of mismatch power injected into power systems; (3d), (3e) and (3f) represent additional regulated outputs produced by different grid-connected units. Particularly, a state transfer model concerning SFR can be actually observed, extracted and illustrated as (3) from (3b) - (3c).

Previous studies generally follow a free residual assumption and confirmed that $\Delta f(t)$ can be adequately evaluated as an analytical closed-form form [1] or a time-discretized form [2]. Unlike existing studies mentioned before, BF-SFR takes

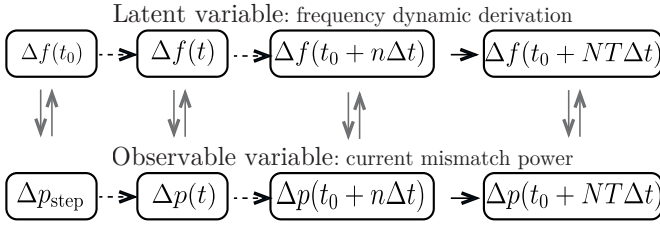


Figure 2. The time-discretized status recurrence process of frequency derivation in the proposed BF-SFR method

account of the impact of UV on frequency dynamics, and the UV problem is abstracted as a sequence of random residuals series. The significance of UV consists of: *i*) the intermittent property of converter-interfaced generators participating in frequency supporting; *ii*) possible measured errors due to the aging or abnormally running of related devices. By treating the SFR with residual assumption as a hidden Markov decision process, the posterior of $\Delta f(t)$ can be estimated in a recursion mathematical form. Let $\mathbb{P}(\Delta f(t_0 + n\Delta t), \mu_1, \sigma_1)$ represents the posterior of $\Delta f(t_0 + n\Delta t)$ at the n -th discretized step; $\mathbb{P}(\Delta p(t_0 + n\Delta t), \mu_2, \sigma_2)$ denotes the posterior of $\Delta p(t_0 + n\Delta t)$ at the n -th discretized step. $\{\mu_1, \sigma_1\}$, $\{\mu_2, \sigma_2\}$ describes the expectation and variance of UV residuals both in the observation process and the measured process. Furthermore, define $\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + (n+1)\Delta t), u_1, \sigma_1)$ to denote an conditional distribution of $\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t), u_1, \sigma_1)$. It describes that the posterior of $\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t), u_1, \sigma_1)$ can be fully inferred after obtaining all measured sequences $\Delta p(t_0 : t_0 + (n+1)\Delta t) = \{\Delta p_t | t \in [t_0, t_0 + (n+1)\Delta t]\}$. its mathematical formulation can be derived as (4) according to the Bayesian theorem.

$\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + (n+1)\Delta t), u_1, \sigma_1)$ in (4) can be indirectly calculated by estimating its prior form $\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + n\Delta t), u_1, \sigma_1)$ and a related normalized constant $\mathbb{P}(\Delta p(t_0 + (n+1)\Delta t) | \Delta f(t_0 + (n+1)\Delta t), u_2, \sigma_2)$. The below gives the mathematical form of the prior function and normalized constant.

$$\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + n\Delta t), u_1, \sigma_1) = \quad (5a)$$

$$\int \{\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta f(t_0 + n\Delta t), \mu_1, \sigma_1) \mathbb{P}(\Delta f(t_0 + n\Delta t) | \Delta p(t_0 : t_0 + n\Delta t), u_1, \sigma_1)\} \cdot d\Delta f(t_0 + n\Delta t)$$

$$\mathbb{P}(\Delta p(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + n\Delta t), u_2, \sigma_2) = \quad (5b)$$

$$\int \{\mathbb{P}(\Delta p(t_0 + (n+1)\Delta t) | \Delta f(t_0 + (n+1)\Delta t), u_2, \sigma_2) \mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + n\Delta t), u_2, \sigma_2)\} d\Delta f(t_0 + (n+1)\Delta t)$$

For the note of convenience, define the initial time $t_0 = 0$ and replace $\langle \Delta f(t_0 + n\Delta t), \Delta p(t_0 + n\Delta t) \rangle$ with their abbreviations

viated forms $\langle \Delta f(n), \Delta p(n) \rangle$, (5) can be further simplified as follows,

$$\mathbb{P}(\Delta f(n+1) | \Delta p(0 : n), u_1, \sigma_1) = \quad (6a)$$

$$\int \{\mathbb{P}(\Delta f(n+1) | \Delta f(n), u_1, \sigma_1) \mathbb{P}(\Delta f(n) | \Delta p(0 : n), u_1, \sigma_1)\} d\Delta f(n+1)$$

$$\mathbb{P}(\Delta p(n+1) | \Delta p(0 : n), u_2, \sigma_2) = \quad (6b)$$

$$\int \{\mathbb{P}(\Delta p(n+1) | \Delta f(n+1), u_2, \sigma_2) \mathbb{P}(\Delta f(n+1) | \Delta p(0 : n), u_2, \sigma_2)\} d\Delta f(n+1)$$

To calculate the posterior function of $\mathbb{P}(\Delta f(n+1) | \Delta p(0 : n), u_1, \sigma_1)$ and $\mathbb{P}(\Delta p(n+1) | \Delta p(0 : n), u_2, \sigma_2)$, the operator $\int(\cdot)$ in (5) can be approximately estimated by an expectation of a certain number of sampled data (or called *filters*), i.e., $\int \varpi[\Delta f(n)] d\Delta f(n) \approx \frac{1}{N} \sum_{n=1}^N \varpi(\lambda_n) \lambda_n$ where ϖ enforces associated integrand function in (6) and λ_n represents possible *filters* generated from the prior of $\Delta f(n)$. Hence, (6) can be rewritten as,

$$\mathbb{P}(\Delta f(n+1) | \Delta p(0 : n), u_1, \sigma_1) =$$

$$1/N \sum_{n=1}^N \mathbb{P}(\Delta f(n+1) | \Delta f(n), u_1, \sigma_1) \lambda'_n \quad (7a)$$

$$\lambda'_n \sim \mathbb{P}(\Delta f(n) | \Delta p(0 : n), u_1, \sigma_1) \quad (7b)$$

$$\mathbb{P}(\Delta p(n+1) | \Delta p(0 : n), u_2, \sigma_2) =$$

$$1/N \sum_{n=1}^N \mathbb{P}(\Delta p(n+1) | \Delta f(n+1), u_2, \sigma_2) \lambda''_n \quad (7c)$$

$$\lambda''_n \sim \mathbb{P}(\Delta f(n+1) | \Delta p(0 : n), u_2, \sigma_2) \quad (7d)$$

where λ' , λ''_n , respectively, represent random filters following associated integrand functions in (7b) and (7d). In combination with (7a) and (7c), the posterior probabilistic assessment of $\Delta f(n+1)$ concerning $\mathbb{P}(\Delta f(n+1))$ can be carried out with the below steps:

- (i) Bayesian inference in the prior stage, here, (7a) is used to generate a number of filters to capture the likely value of $\Delta f(n+1)$;
- (ii) Bayesian predication in the prediction stage, (4) and (7c) is used to update the posterior of $\Delta f(n+1)$. Since λ' , λ''_n cannot directly generated from $\mathbb{P}(\Delta f(n) | \Delta p(0 : n), u_1, \sigma_1)$ and $\mathbb{P}(\Delta f(n+1) | \Delta p(0 : n), u_2, \sigma_2)$ because their functions are unknown, the sequential important sampling (SIS) algorithm is applied to estimate the value of studied integrand functions. Specially, the solution of the i -th prior stage concerning $\mathbb{P}(\Delta f(n) | \Delta p(0 : n), u_1, \sigma_1)$ is took as an example to describe the solution process via SIS, another case of the later (posterior) stage concerning $\mathbb{P}(\Delta f(n+1) | \Delta p(0 : n), u_2, \sigma_2)$ is in a similar solution.

Let $\iota(n) = \mathbb{P}(\Delta f(n) | \Delta p(0 : n), u_2, \sigma_2)$, $\iota(n) = (\iota(n)/\lambda(n)) \lambda(n)$ where $\iota(n)$ is the original probabilistic density function of posterior distribution at $t=n\Delta t$; $\lambda(n)$ enforces a proposed function. $\gamma(n) = \iota(n)/\lambda(n)$ represents a likelihood coefficient.

$$\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + (n+1)\Delta t), u_1, \sigma_1) = \quad (4)$$

$$\frac{\mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + n\Delta t), u_1, \sigma_1) \mathbb{P}(\Delta p(t_0 + (n+1)\Delta t) | \Delta f(t_0 + (n+1)\Delta t), u_1, \sigma_1)}{\mathbb{P}(\Delta p(t_0 + (n+1)\Delta t) | \Delta p(t_0 : t_0 + n\Delta t), u_2, \sigma_2)}$$

Table I. The simulation of BF-SFR through SIS algorithm

NO.	Calculation processes
1	Initialize: Discretized step $\Delta t = 0.05s$; Initialized time $t_0 = 0s$ Counter $n = 0$; Filter number $NT = 1000$
2	For $n = 0$, Then
3	Let $t = t_0 + \Delta t \times n$, Then
4	Inference process: Generating possible (instant) filters $(\Delta f)^{(n)}(t_0 + (n+1)\Delta t)$ from proposed distribution: $\lambda'(n+1) \sim \mathbb{P}(\Delta f(t_0 + (n+1)\Delta t) \Delta p(t_0 : t_0 + n\Delta t), \mu_1, \sigma_1)$ in (6a);
5	Prediction process: Generating possible observation filters $(\Delta p)^{(n)}(t_0 + (n+1)\Delta t)$ from normalized distribution: $\lambda''(n+1) \sim \mathbb{P}(\Delta p(t_0 + (n+1)\Delta t) \Delta p(t_0 : t_0 + n\Delta t), \mu_2, \sigma_2)$ in (6b);
6	Updating weight coefficients: Calculating γ' and γ'' through (9), and Re-estimate original filters $(\Delta f)^{(n)}(t_0 + (n+1)\Delta t)$ via (4)
7	End Let
8	End For

Hence, the *filters* following $\iota(n)$ can be alternatively sampled from $\lambda(n)$ with a related weight coefficient $\gamma(n)$, i.e., $\iota(n) = \lambda(n)\gamma(n)$.

$$\gamma(n+1) = \frac{\iota(\Delta f(n+1) | \Delta p(0:n+1), \mu_1, \sigma_1)}{\lambda(\Delta f(n+1) | \Delta p(0:n+1), \mu_2, \sigma_2)} = \gamma(n) \times \left(\frac{\iota(\Delta f(n+1) | \Delta f(t), \mu_1, \sigma_1) \iota(\Delta p(n+1) | \Delta f(n+1), \mu_2, \sigma_2)}{\lambda(\Delta f(n+1) | \Delta f(t), \mu_1, \sigma_1)} \right) \quad (8)$$

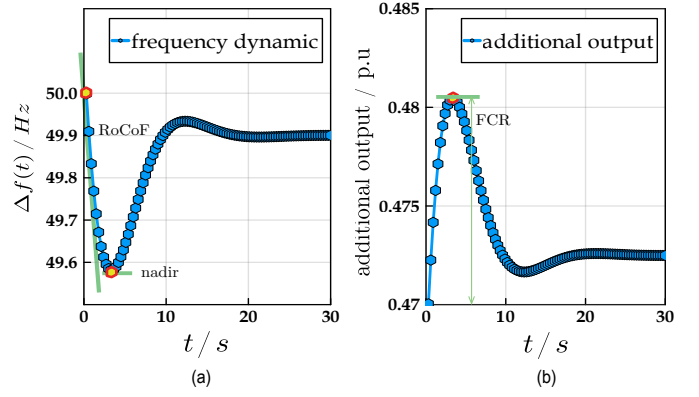
Substituting $\iota(n+1) = \lambda(n+1)\gamma(n+1)$, $\iota(n) = \lambda(n)\gamma(n)$ into (4), $\iota(n+1)$ can be recursively calculated through (8). In (8), another critical issue to be reviewed is the selection of the proposed distribution. Here, the recursive transfer distribution $\iota(\Delta f(n+1) | \Delta f(n), \mu_1, \sigma_1)$ can be availability used to simplify the calculation process. Accordingly, the updating weight (8) can be reduced as,

$$\gamma(n+1) = \gamma(n) \iota(\Delta p(n+1) | \Delta f(n+1), \mu_2, \sigma_2) \quad (9)$$

In (9), the likelihood rate in each iteration can be recursively derived from $\iota(\Delta p(n+1) | \Delta f(n+1), \mu_2, \sigma_2)$. On this basis, let $(\Delta f)^{(n)}(t_0 + (n+1)\Delta t)$, $(\Delta p)^{(n)}(t_0 + (n+1)\Delta t)$ define the sampled *filters* from the pro-selected proposed function $\lambda'(n+1)$ in the inference process and $\lambda''(n+1)$ in the predication process, respectively. Likewise, let $\gamma'(n+1)$, $\gamma''(n+1)$, respectively, represent the likelihood rate at time $t = t_0 + (n+1)\Delta t$ both in Bayesian-filter-based calculation processes. Table I describes the Pseudo-Code of the proposed BF-SFR method.

III. FREQUENCY CONSTRAINT REFORMULATION

In this section, the significant requirements of SFR including *RoCoF*, *nadir* and *FCR* are investigated and reformulated a set of related constraints incorporating into UC studies. The physical meaning of these feature frequency requirements are illustrated in Fig. 6. It is worth noting that the reconstructed chance-constrained frequency nadir constraint is based on the new BF-SFR model in section II, as well as other frequency-related constraints (that is, *RoCoF* and *FCR*) can be referred in [6-8].

**Figure 3.** Three critical frequency dynamic requirements of power systems: *RoCoF*, *nadir* and *FCR*

A. *RoCoF* constraint

To prevent $\Delta f(t)$ from changing too fast at inertia response, the differential of frequency derivation $\Delta f(t)$, denoted by $\Delta f'(t)$, is included, i.e., $\Delta f'(t) \leq RoCoF_{max}$. Since $\Delta f'(t) = \Delta p_{step} / M_{equ} : M_{equ} = M_G + M_W$, the *RoCoF* constraint can be obtained,

$$\Delta p_{step} / M_{equ} \leq RoCoF_{max} \quad (10)$$

$$\Leftrightarrow M_G \geq \Delta p_{step} / RoCoF_{max} - M_W, \forall g, t$$

$$\Leftrightarrow \sum_{g=1}^{NG} M_g x_{g,t} \geq \Delta p_{step} / RoCoF_{max} - \sum_{w=1}^{NW} M_w, \forall g, t$$

B. *FCR* constraint

To guarantee adequate frequency reserve provided by grid-connected generators, both conventional generator, converter-based generators, and BESS must remain available frequency containment reserve against imbalanced disturbances. In combination with (3), the additional power $\langle \Delta p_{add}^G, \Delta p_{add}^W, \Delta p_{add}^E \rangle$ of different units can be solved. To keep these grid-connected units have adequate *FCR*, the lower bound of *FCR* concerning different units can be derived, that is, $\sum_{g \in G} p_{g,s,t} \geq \Delta p_{add}^G(t)$; $\sum_{w \in W} p_{w,s,t} \geq \Delta p_{add}^W(t)$ and $\sum_{e \in E} p_{e,s,t} \geq \Delta p_{add}^E(t)$, $\forall t, s$. By replacing $\Delta p_{add}^G(t)$, $\Delta p_{add}^W(t)$, and $\Delta p_{add}^E(t)$ with their analytical form in (3d), (3e) and (3f), another equivalent constraint concerning *FCR* can be derived,

$$\sum_{w=0}^{NW} (p_w^{max} - p_{w,s,t}) \geq R_W^{-1} \Delta f(t) \quad (11a)$$

$$\sum_{e=0}^{NE} (p_e^{max} - p_{e,s,t}^{dis}) \geq R_E^{-1} \Delta f(t) \quad (11b)$$

$$\sum_{g=0}^{NG} (p_g^{max} - p_{g,s,t}) \geq K_G R_G^{-1} \Delta f(t_0 + n\Delta t) - K_G (1 - F_G)$$

$$R_G^{-1} e^{-\frac{1}{T_G}(t_0 + n\Delta t)} \left(\sum_{i=0}^n e^{\frac{1}{T_G}(t_0 + i\Delta t)} \Delta f'(t_0 + i\Delta t) \right) \Delta t \quad (11c)$$

Note that the upper bound of $\Delta f(t)$ in (11) is predefined as Δf_{max} , therefore, a concise form of (11a) and (11b) can be calculated by replacing $\Delta f(t)$ with Δf_{max} ,

$$\sum_{w=0}^{NW} (p_w^{max} - p_{w,s,t}) \geq \Delta f_{max} / R_W, \forall s, t \quad (12a)$$

$$\sum_{e=0}^{NE} (p_e^{\max} - p_{e,s,t}^{\text{dis}}) \geq \Delta f_{\max}/R_E, \forall s, t \quad (12b)$$

In the context of the *FCR* constraint (11c) of grid-connected conventional generators, the right-hand side of (11c), representing adjustable output of conventional generators, can be rewritten as,

$$\begin{aligned} & K_G R_G^{-1} \Delta f(t_0 + n\Delta t) - (K_G(1 - F_G) R_G^{-1} \text{RoCoF}_{\max}) (n\Delta t) \\ & \leq K_G R_G^{-1} \Delta f(t_0 + n\Delta t) - (K_G(1 - F_G) R_G^{-1} \text{RoCoF}_{\max}) \\ & \quad \left(e^{-\frac{1}{T_G}(t_0+n\Delta t)} \sum_{i=0}^n e^{\frac{1}{T_G}(t_0+i\Delta t)} \right) \Delta t \\ & \leq K_G R_G^{-1} \Delta f(t_0 + n\Delta t) - K_G(1 - F_G) R_G^{-1} \times \\ & \quad e^{-\frac{1}{T_G}(t_0+n\Delta t)} \left(\sum_{i=0}^n e^{\frac{1}{T_G}(t_0+i\Delta t)} \Delta f'(t_0 + i\Delta t) \right) \Delta t \\ & \leq K_G R_G^{-1} \Delta f(t_0 + n\Delta t) \leq K_G R_G^{-1} \Delta f_{\max} \end{aligned} \quad (13)$$

By replacing the upper bound of adjustable output in (13) into (11c), a slacked *FCR* constraint concerning conventional generators can be derived:

$$\sum_{g=1}^{NG} (p_g^{\max} - p_{g,s,t}) \geq K_G R_G^{-1} \Delta f_{\max}, \forall s, t \quad (14)$$

C. (chance – constraint) nadir constraint generation and reformulation

For preventing $\Delta f(t)$ greater than its predefined threshold $\Delta f_{\max}$, the possible fluctuation of $\Delta f_{\text{nadir}} = \max\{\Delta f(t)\}$ needs to be further limited, i.e., $\Delta f_{\text{nadir}} \leq \Delta f_{\max}$. Notice that both statistical expectation and probabilistic density distribution of $\Delta f(t) : t \in \{t_0, t_0 + n\Delta t\}$ can be successfully solved through BF-SFR, a chance-constrained *nadir* constraint could be constructed as below,

$$\mathbb{P}(\Delta f_{\text{nadir}} \leq \Delta f_{\max}) \geq 1 - \epsilon \quad (15)$$

where ϵ is the confidence risk level associated with the chance constraint (15); Δf_{nadir} is closely dependent with the initial imbalance disturbance Δp_{step} , aggregated frequency parameters of various grid-connected units $\langle M_G, D_G, R_G, K_G, T_G, R_W, R_E \rangle$ and external UV characteristic $\langle \mu_1, \sigma_1, \mu_2, \sigma_2 \rangle$. For simplification purpose, let $\wp = \langle M_G, D_G, R_G, K_G, T_G, R_W, R_E \rangle$ and $\zeta = \langle \mu_1, \sigma_1, \mu_2, \sigma_2 \rangle$, and then, Δf_{nadir} can be rewritten a nonlinear mapping function $\Psi(\wp, \zeta)$, i.e., $\Delta f_{\text{nadir}} = \Psi(\wp, \zeta)$. Furthermore, a sampled average approximate (SAA) method is used to construct the deterministic convex formulation of (15) which can be readily identified by any off-the-shelf MIP solvers so as to subsequently integrate it into the baseline UC studies. More detailed, by introducing a set of binary indicator variable $z_q : q \in \{1, 2, \dots, NQ\}$ and using the Big-M method, (15) can be equivalently reformulated with a set of linearized constraints as follows,

$$\Psi(\wp, \hat{\zeta}^{(q)}) \leq \Delta f_{\max} + M z_q, \forall q \quad (16a)$$

$$\sum_{q=1}^{NQ} \pi_q z_q \leq \epsilon, \forall q \quad (16b)$$

where (16) represents an approximately mix-integer linearized constraints of (15). $z_q = 0$ indicates an active constraint where the *nadir* must be required for the predefined uncertain variability condition; $z_q = 1$ describes an inactive constraint where

the (15) at the predefined sampled uncertain condition q could be ignored. Accordingly, the redesigned chance constraint (16) has been reshaped as an equivalence of restricting the number of $z_q, 1 \leq q \leq NQ$ to be ones. It is important to mention that the mathematical reformulation of the similar type constraint (15) are not limited to the form (16). By directly introducing z_q but no longer introducing M , a bi-linear type mixed integer reformulation of (15) is also feasible, that is,

$$\left(\Psi(\wp, \hat{\zeta}^{(q)}) - \Delta f_{\max} \right) (1 - z_q) \leq 0, \forall q \quad (17a)$$

$$\sum_{q=1}^{NQ} \pi_q z_q \leq \epsilon, \forall q \quad (17b)$$

The expansion of (17a) can be rewritten as: $\Psi(\wp, \hat{\zeta}^{(q)}) - z_q \Psi(\wp, \hat{\zeta}^{(q)}) - (1 - z_q) \Delta f_{\max} \leq 0, \forall q$ where the nonlinear term $\Psi(\wp, \hat{\zeta}^{(q)})$ needs to be piece-wise-linearized (PWL) and the bi-linear one $z_q \Psi(\wp, \hat{\zeta}^{(q)})$ could be further reformulated through McCormick linearization method. For pretending the resulting constraint from being excessive complexity, (16) is utilized to approximate (15). Note that (16a) cannot be directly incorporated into UC due to the nonlinear characteristic of $\Psi(\wp, \hat{\zeta}^{(q)})$, thereby, a “max-affine” PWL method is implemented to estimate the original value of $\Psi(\wp, \hat{\zeta}^{(q)})$.

Let $\hat{\Psi}(\wp, \hat{\zeta}^{(q)}) = \max \left(\alpha_{\iota,1}^{(q)} \tilde{H}_{\text{equ}}^{(q)} + \alpha_{\iota,2}^{(q)} \tilde{D}_{\text{equ}}^{(q)} + \alpha_{\iota,3}^{(q)} \left(\frac{\tilde{F}_G^{(q)}}{\tilde{R}_G^{(q)}} \right) + \alpha_{\iota,4}^{(q)} \right)$ describe the approximated fitting function of $\hat{\Psi}(\wp, \hat{\zeta}^{(q)})$ given a stochastic uncertain realization $\hat{\zeta}^{(q)}$; $\langle \tilde{H}_{\text{equ}}^{(q)}, \tilde{D}_{\text{equ}}^{(q)}, \tilde{F}_G^{(q)}, \tilde{R}_G^{(q)} \rangle$ denotes a possible set of aggregated frequency parameters with the respect of $\hat{\zeta}^{(q)}$. In such a situation, $\hat{\Psi}(\wp, \hat{\zeta}^{(q)})$ can be accessed by observing the maximum value of constructed hyperplane $\Theta_{\iota}^{(q)} : \iota \in \{1, 2, 3, 4\}$ where $\Theta_{\iota}^{(q)} = \alpha_{\iota,1}^{(q)} \tilde{H}_{\text{equ}}^{(q)} + \alpha_{\iota,2}^{(q)} \tilde{D}_{\text{equ}}^{(q)} + \alpha_{\iota,3}^{(q)} \left(\frac{\tilde{F}_G^{(q)}}{\tilde{R}_G^{(q)}} \right)$. Here, the selection of fitting parameters $\langle \alpha_{\iota,1}^{(q)}, \alpha_{\iota,2}^{(q)}, \alpha_{\iota,3}^{(q)}, \alpha_{\iota,4}^{(q)} \rangle$ can be solved by minimizing the lose function defined by $\vartheta_{\text{loss}} = \hat{\Psi}(\wp, \hat{\zeta}^{(q)}) - \Psi(\wp, \hat{\zeta}^{(q)})$. As such, the below lists the detailed model on calculating fitting parameters.

$$\begin{aligned} \min \sum_{r=1}^{NR} & \left(\max \left(\alpha_{\iota,1}^{(q)} \tilde{H}_{\text{equ}}^{(q)} + \alpha_{\iota,2}^{(q)} \tilde{D}_{\text{equ}}^{(q)} + \alpha_{\iota,3}^{(q)} \left(\frac{\tilde{F}_G^{(q)}}{\tilde{R}_G^{(q)}} \right) + \alpha_{\iota,4}^{(q)} \right) \right. \\ & \left. - \Psi(\wp, \hat{\zeta}^{(q)}) \right)^2, \forall \iota \in \{1, 2, 3\} \end{aligned} \quad (18)$$

In (18), the inner “max” operator automatically chooses the most suitable hyperplane closet to each evaluation point, and the outer “min” operator guarantees the global accuracy between preselected hyperplanes and all evaluation points. In the term of problem solving, (18) can be solved by directly calling Ipopt solver or reshaped as its recursive form and subsequently calling existing MIP solver, e.g., Gurobi, IBM CPLEX, etc. More detailed on the calculation of (18) can be found in [].

Substituting $\hat{\Psi}(\wp, \hat{\zeta}^{(q)})$ through (18) into (16a), a tractable convex form of (16a) can be obtained as below,

$$\begin{aligned} & \alpha_{\iota,1}^{(q)} \tilde{H}_{\text{equ}}^{(q)} + \alpha_{\iota,2}^{(q)} \tilde{D}_{\text{equ}}^{(q)} + \alpha_{\iota,3}^{(q)} \left(\frac{\tilde{F}_G^{(q)}}{\tilde{R}_G^{(q)}} \right) + \alpha_{\iota,4}^{(q)} \\ & \leq \Delta f_{\max} + M z_q, \forall q, \iota \in \{1, 2, 3, 4\} \end{aligned} \quad (19a)$$

IV. CHANCE-CONSTRAINED FREQUENCY DYNAMIC UNIT COMMITMENT (CCFDUC)

This section gives a complete formulation of the proposed CCFDUC model in this work. By incorporating frequency dynamic constraints in section III into the baseline UC problem, the mathematical form of the proposed UC model can be obtained as below,

$$\min \sum_{t=1}^{NT} \left[\sum_{g=1}^{NG} (c_g^{\text{up}} u_{g,t} + c_g^{\text{dw}} u_{g,t} + c_g^{\text{lt}} u_{g,t}) + \right. \quad (20a)$$

$$\left. \sum_{s=1}^{NS} \pi_s \left(\sum_{g=1}^{NG} \sum_{z=1}^{NZ} F_g^z(p_{g,s,t}^{(z)}) + e^{\text{voLL}} \sum_{d=1}^{ND} p_{d,s,t} \right) \right], \forall t, g, s, w, d$$

$$\sum_{m=\lceil t-T_g^{\text{su}}+1,0 \rceil}^t u_{g,m} \leq x_{g,t}, \forall g, t \quad (20b)$$

$$\sum_{m=\lceil t-T_g^{\text{sd}}+1,0 \rceil}^t u_{g,m} \leq 1 - x_{g,t}, \forall g, t \quad (20c)$$

$$x_{g,t} - x_{g,t-1} = u_{g,t} - v_{g,t}, \forall g, t \quad (20d)$$

$$u_{g,t} + v_{g,t} \leq 1, \forall g, t \quad (20e)$$

$$\sum_{g=1}^{NG} p_{g,s,t} + \sum_w^{NW} (p_{w,s,t} - \Delta p_{w,s,t}) + \sum_{e=1}^{NE} (p_{e,s,t}^{\text{dis}} - \Delta p_{e,s,t}^{\text{ch}}) -$$

$$\sum_{d=1}^{ND} (p_{d,t} - \Delta p_{d,s,t}) = 0, \forall g, w, e, s, t \quad (20f)$$

$$\left| \sum_{b=1}^{NB} \kappa_{l,b} \left(\sum_{g=1}^{NG_b} p_{g,s,t} + \sum_{w=1}^{NW_b} (p_{w,s,t} - \Delta p_{w,s,t}) + \sum_{e=1}^{NE_b} (p_{e,s,t}^{\text{dis}} - \Delta p_{e,s,t}^{\text{ch}}) - \sum_{d=1}^{ND} (p_{d,t} - \Delta p_{d,s,t}) \right) \right| \leq p_i^{\text{max}}, \forall t, l, s \quad (20g)$$

$$p_{g,s,t} - p_{g,s,t-1} \leq \gamma_g^{\text{ru}} x_{g,t} + p_g^{\text{max}} (1 - x_{g,t}) + \gamma_g^{\text{su}} (x_{g,t} - x_{g,t-1}), \forall g, s, t \quad (20h)$$

$$p_{g,s,t-1} - p_{g,s,t} \leq \gamma_g^{\text{rd}} x_{g,t} + p_g^{\text{max}} (1 - x_{g,t-1}) + \gamma_g^{\text{sd}} (x_{g,t-1} - x_{g,t}), \forall g, s, t \quad (20i)$$

$$\sum_{g=1}^{NG} (x_{g,t} p_g^{\text{max}} - p_{g,s,t}) + \sum_{e=1}^{NE} (p_{e,s,t}^{\text{dis,max}} - p_{e,s,t}^{\text{dis}}) \geq$$

$$\varrho_1^+ \sum_{d=1}^{ND} p_{d,t} + \varrho_2^+ \sum_{e=1}^{NE} p_{w,s,t}^{\text{max}}, \forall s, t \quad (20j)$$

$$\sum_{g=1}^{NG} (p_{g,s,t} - x_{g,t} p_g^{\text{min}}) + \sum_{e=1}^{NE} (p_{e,s,t}^{\text{ch,max}} - p_{e,s,t}^{\text{ch}}) \geq$$

$$\varrho_1^- \sum_{d=1}^{ND} p_{d,t} + \varrho_2^- \sum_{e=1}^{NE} p_{w,s,t}^{\text{max}}, \forall s, t \quad (20k)$$

$$x_{g,t} p_g^{\text{max}} \leq p_{g,s,t} \leq x_{g,t} p_g^{\text{min}}, \forall s, t \quad (20l)$$

$$0 \leq \Delta p_{w,s,t} \leq p_{w,s,t}^{\text{max}}, \forall w, s, t \quad (20m)$$

$$0 \leq \Delta p_{d,s,t} \leq p_{d,t}, \forall d, s, t \quad (20n)$$

$$0 \leq p_{e,s,t}^{\text{dis}} \leq u_{e,s,t}^{\text{dis}} p_e^{\text{dis,max}}, \forall e, s, t \quad (20o)$$

$$0 \leq p_{e,s,t}^{\text{ch}} \leq u_{e,s,t}^{\text{ch}} p_e^{\text{ch,max}}, \forall e, s, t \quad (20p)$$

$$q_{e,s,t+1} = q_{e,s,t} + (\eta_e^{\text{ch}} p_{e,s,t}^{\text{ch}} - \eta_e^{\text{dis}} p_{e,s,t}^{\text{dis}}) \Delta t, \forall e, s, t \quad (20q)$$

$$q_{e,s,t=1} = q_{e,s,t=NT}, \forall e, s, t \quad (20r)$$

$$u_{e,s,t}^{\text{dis}} + u_{e,s,t}^{\text{ch}} \leq 1, \forall e, s, t \quad (20s)$$

$$p_{g,s,t} = \sum_{z=1}^{NZ} p_{g,s,t}^{(z)}, \forall z, g, s, t \quad (20t)$$

$$0 \leq p_{g,s,t}^{(z)} \leq x_{g,t} p_{g,s,t}^{\text{max}}, \forall z, g, s, t \quad (20u)$$

$$F_g^z(p_{g,s,t}^{(z)}) \geq k_g^{(z)} p_{g,s,t}^{(z)} + x_{g,t} \omega_g^{(z)}, \forall z, g, s, t \quad (20v)$$

$$\sum_{g=1}^{NG} M_g x_{g,t} \geq \Delta p_{\text{step}} / RoCoF_{\text{max}} - \sum_{w=1}^{MW} M_w, \forall g, w, t \quad (21a)$$

$$\sum_{g=1}^{NG} (p_g^{\text{max}} - p_{g,s,t}) \geq K_G R_G^{-1} \Delta f_{\text{max}}, \forall s, t \quad (21b)$$

$$\sum_{w=0}^{NG} (p_w^{\text{max}} - p_{w,s,t}) \geq \Delta f_{\text{max}} / R_W, \forall s, t \quad (21c)$$

$$\sum_{e=0}^{NE} (p_e^{\text{max}} - p_{e,s,t}^{\text{dis}}) \geq \Delta f_{\text{max}} / R_E, \forall s, t \quad (21d)$$

$$\sum_{q=1}^{NQ} \pi_q \phi_{q,t} \leq \epsilon, \forall q, t \quad (21e)$$

$$\alpha_{\iota,1}^{(q)} H_{\text{equ},t} + \alpha_{\iota,2}^{(q)} D_{\text{equ},t} + \alpha_{\iota,3}^{(q)} (F_{G,t} / R_{G,t}) + \alpha_{\iota,4}^{(q)} \leq \Delta f_{\text{max}} + M \phi_{q,t}, \forall q, \iota \in \{1, 2, 3, 4\} \quad (21f)$$

$$M_{\text{equ},t} = p_{\text{base}}^{-1} \left(\sum_{g=1}^{NG} (M_g p_g^{\text{max}}) x_{g,t} + \sum_{w=1}^{NW} (M_w p_{w,s,t}) \right) \quad (21g)$$

$$D_{\text{equ},t} = p_{\text{base}}^{-1} \left(\sum_{g=1}^{NG} (D_g p_g^{\text{max}}) x_{g,t} + \sum_{w=1}^{NW} (R_w^{-1} p_{w,s,t}) + \sum_{e=1}^{NE} (R_e^{-1} p_e^{\text{max}}) \right), \forall t \quad (21h)$$

$$F_{G,t} = p_{\text{base}}^{-1} \sum_{g=1}^{NG} (K_g F_g R_g^{-1} p_g^{\text{max}}) x_{g,t}, \forall t \quad (21i)$$

$$R_{G,t}^{-1} = p_{\text{base}}^{-1} \sum_{g=1}^{NG} (K_g R_g^{-1} p_g^{\text{max}}) x_{g,t}, \forall t \quad (21j)$$

$$R_{W,t}^{-1} = p_{\text{base}}^{-1} \sum_{w=1}^{NW} (K_w R_w^{-1} p_{w,s,t}), \forall t \quad (21k)$$

$$R_{E,t}^{-1} = p_{\text{base}}^{-1} \sum_{e=1}^{NE} (K_e R_e^{-1} p_e^{\text{max}}), \forall t \quad (21l)$$

$$x_{g,t} \in \{0, 1\}, u_{g,t} \in \{0, 1\}, v_{g,t} \in \{0, 1\}, u_{e,s,t}^{\text{dis}} \in \{0, 1\}, u_{e,s,t}^{\text{ch}} \in \{0, 1\} \quad (21m)$$

$$p_{g,s,t} \in \mathbb{R}^+, \Delta p_{w,s,t} \in \mathbb{R}^+, \Delta p_{d,s,t} \in \mathbb{R}^+, p_{g,s,t}^{\text{dis}} \in \mathbb{R}^+,$$

$$p_{g,s,t}^{\text{ch}} \in \mathbb{R}^+, q_{e,s,t} \in \mathbb{R}^+ \quad (21n)$$

In the above formulation, (20a)-(20v) define a baseline UC model, (21a)-(21l) denote frequency dynamic constraints in line with the actual frequency requirements. It worth noting that the newly redesigned *nadir* constraint (21e)-(21f) is based

on BF-SFC which is significantly different with previous SFR model or UC studies.

Constraint (20a) denotes the objective of the UC problem, it aims to minimize the total cost of conventional generators cost. Constraint (20b) and (20c), respectively, declare the minimum uptime and downtime restriction of all conventional generators. Constraint (20d) and (20e) respectively guarantee the consistency of shut-up/shut-down decision and operation status of individual conventional generator. Constraint (20f) describes (active) power balance restriction over the entire dispatching periods. Constraint (20g) indicates each transmission line's capacity restriction. Constraints (20h) and (20i) respectively guarantee the up-forward and down-forward ramping rate of each conventional generator between adjacent dispatching periods. Constraints (20j) and (20k) describe the requirement of up-forward and down-forward spinning reserve capacity produced by each conventional generator. Constraint (20l) indicate the minimize/minimize power of multiple generators. Constraint (20m) and (20n) impose restrictions on abandoned wind output and forced load curtailment. Constraint (20o) and (20p), respectively, declare the minimize/maximize capacities of charging/discharging output with respect to each BESS. Constraint (20q) represents a sequential coupling limitation between BESS energy and related discharging/charging decisions. Constraint (20r) defines a periodic energy regulation policy for each BESS. Constraint (20s) emphasizes the unique status of discharging/charging decisions of individual BESS. Constraint (20t) imposes individual conventional generator's output as the sum of the segmented output generated on each block of associated linearized fuel cost curve. Constraint (20u) enforces the binding restriction of each segmented output. Constraint (20v) declares the operation cost of individual conventional generator. Constraint (21a), (21b)-(21d), and (21e)-(21f) respectively denote *RoCoF*, *FCR* and *nadir* constraint against selected power disturbance. It is worth noting that the original chance constraint representing *nadir* bindings in (15) has been reshaped a set of convex constraint through SAA. Constraints (21i)-(21j) respectively denote aggregated inertia constant, aggregated damping constant, aggregated fraction of total output generated by the high pressure turbine of reheated generators, and aggregated droop coefficient. Constraints (21k) and (21l) denote aggregated droop coefficient of wind farms and BESS. Constraint (21m)-(21n) describe the feasible space of related binary decision variables and continuous decision variables, respectively.

\title{The Title of Your Paper}

Please try to avoid the use of math or chemical formulas in your title if possible.

A. Author Names and Affiliations

The author section should be coded as follows:

```
\author{Masahito Hayashi
\IEEEmembership{Fellow, IEEE}, Masaki Owari
\thanks{M. Hayashi is with Graduate School
of Mathematics, Nagoya University, Nagoya,
Japan}
```

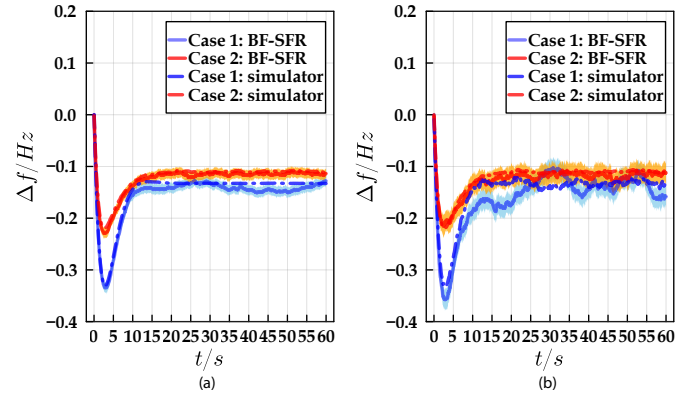


Figure 4. The unified frequency dynamic response of electrical power systems against occurred power disturbance

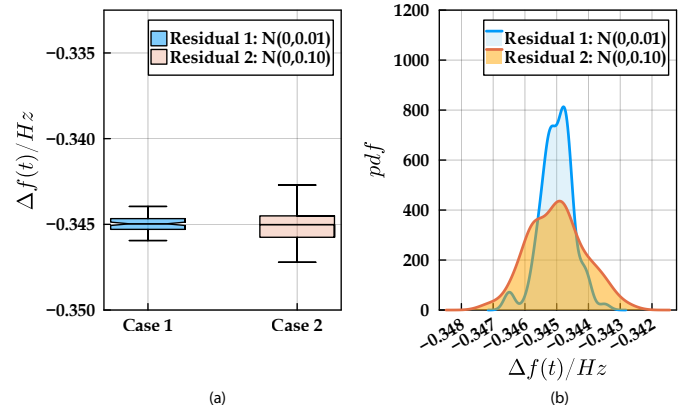


Figure 5. The unified frequency dynamic response of electrical power systems against occurred power disturbance

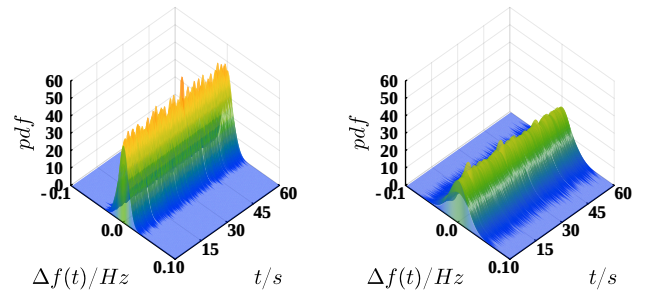


Figure 6. The unified frequency dynamic response of electrical power systems against occurred power disturbance

```
\thanks{M. Owari is with the Faculty of
Informatics, Shizuoka University,
Hamamatsu, Shizuoka, Japan.}
}
```

Be sure to use the \IEEEmembership command to identify IEEE membership status. Please see the “IEEEtran_HOWTO.pdf” for specific information on coding authors for Conferences and Computer Society publications. Note that the closing curly brace for the author group comes at the end of the thanks

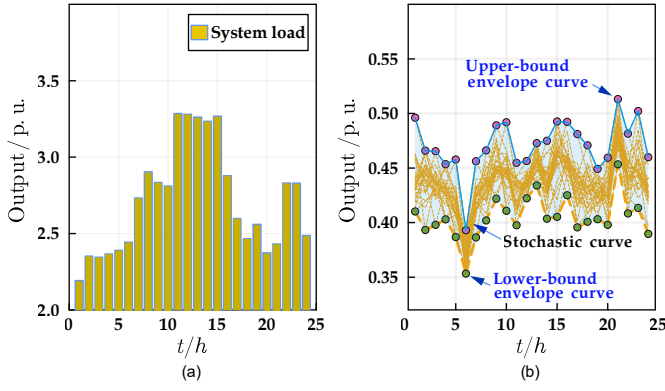


Figure 7. The unified frequency dynamic response of electrical power systems against occurred power disturbance

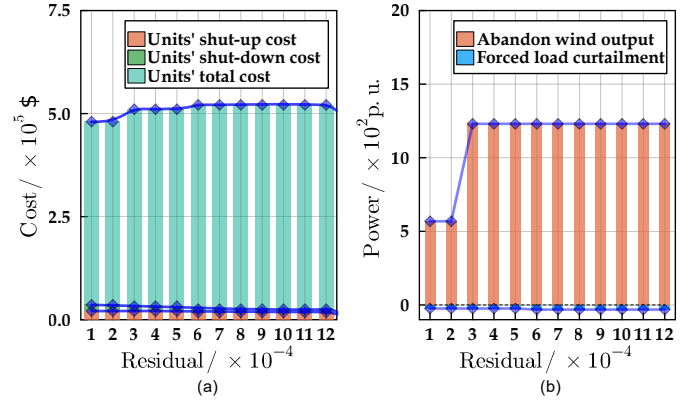


Figure 11. The unified frequency dynamic response of electrical power systems against occurred power disturbance

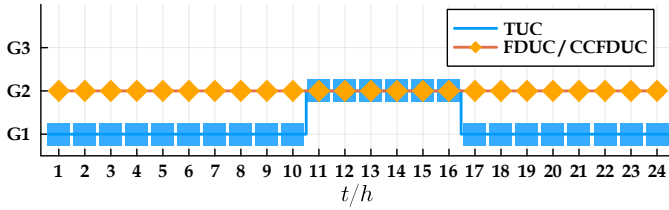


Figure 8. The unified frequency dynamic response of electrical power systems against occurred power disturbance

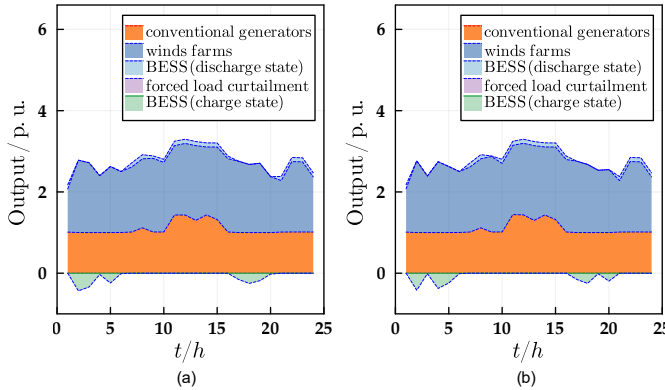


Figure 9. The unified frequency dynamic response of electrical power systems against occurred power disturbance

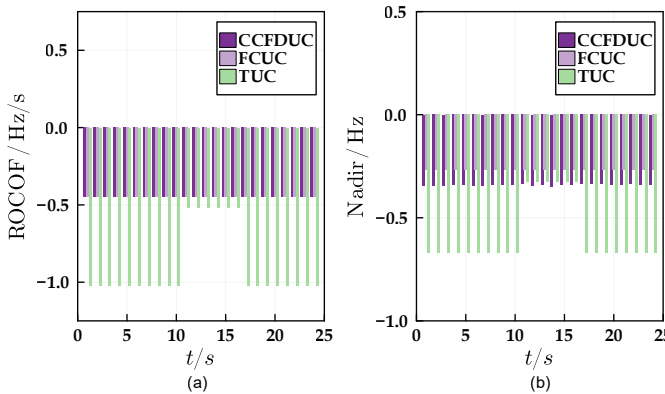


Figure 10. The unified frequency dynamic response of electrical power systems against occurred power disturbance

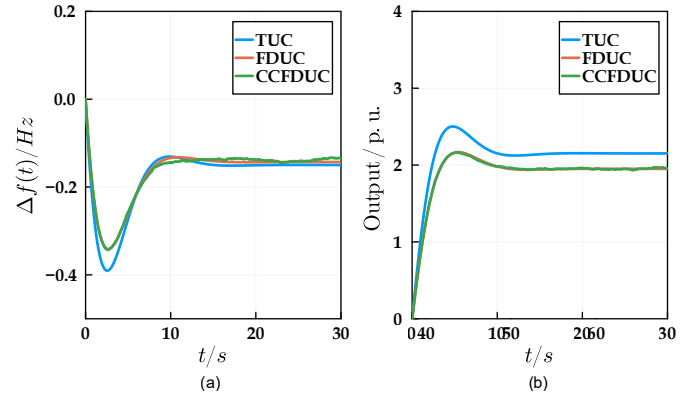


Figure 12. The unified frequency dynamic response of electrical power systems against occurred power disturbance

group. This will prevent you from creating a blank first page.

B. Running Heads

The running heads are declared by using the `\markboth` command. There are two arguments to this command: the first contains the journal name information and the second contains the author names and paper title.

```
\markboth{Journal of Quantum Electronics,
Vol. 1, No. 1, January 2021}
{Author1, Author2,
\MakeLowercase{\textit{(et al.)}}:
Paper Title}
```

C. Copyright Line

For Transactions and Journals papers, this is not necessary to use at the submission stage of your paper. The IEEE production process will add the appropriate copyright line. If you are writing a conference paper, please see the “IEEE-tran_HOWTO.pdf” for specific information on how to code “Publication ID Marks”.

D. Abstracts

The abstract is the first element of a paper after the `\maketitle` macro is invoked. The coding is simply:


```
\begin{abstract}
Text of your abstract.
\end{abstract}
```

Please try to avoid mathematical and chemical formulas in the abstract.

E. Index Terms

The index terms are used to help other researchers discover your paper. Each society may have its own keyword set. Contact the EIC of your intended publication for this list.

```
\begin{IEEEkeywords}
Broad band networks, quality of service
\end{IEEEkeywords}
```

V. HOW TO CREATE COMMON BODY ELEMENTS

The following sections describe common body text elements and how to code them.

A. Initial Drop Cap Letter

The first text paragraph uses a “drop cap” followed by the first word in ALL CAPS. This is accomplished by using the `\IEEEPARstart` command as follows:

```
\IEEEPARstart{T}{his} is the first paragraph
of your paper. . .
```

B. Sections and Subsections

Section headings use standard L^AT_EX commands: `\section`, `\subsection` and `\subsubsection`. Numbering is handled automatically for you and varies according to type of publication. It is common to not indent the first paragraph following a section head by using `\noindent` as follows:

```
\section{Section Head}
\noindent The text of your paragraph . . .
```

C. Citations to the Bibliography

The coding for the citations are made with the L^AT_EX `\cite` command. This will produce individual bracketed reference numbers in the IEEE style. At the top of your L^AT_EX file you should include:

```
\usepackage{cite}
```

For a single citation code as follows:

```
see \cite{ams}
```

This will display as: see [1]

For multiple citations code as follows:

```
\cite{ams,oxford,lacomp}
```

This will display as [1]–[3]

D. Figures

Figures are coded with the standard L^AT_EX commands as follows:

```
\begin{figure}[!t]
\centering
\includegraphics[width=2.5in]{fig1}
\caption{This is the caption for one fig.}
\label{fig1}
\end{figure}
```

The `[!t]` argument enables floats to the top of the page to follow IEEE style. Make sure you include:

```
\usepackage{graphicx}
```

at the top of your L^AT_EX file with the other package declarations.

To cross-reference your figures in the text use the following code example:

```
See figure \ref{fig1} ...
```

This will produce:

See figure 1 . . .

E. Tables

Tables should be coded with the standard L^AT_EX coding. The following example shows a simple table.

```
\begin{table}
\begin{center}
\caption{Filter design equations ...}
\label{tab1}
\begin{tabular}{| c | c | c |}
\hline
Order & Arbitrary coefficients & 
coefficients\\
of filter &  $e_m$  &  $\beta_{ij}$  \\
\hline
1 &  $\beta_{ij} = \hat{e} \cdot \hat{\beta}_{ij}$ , & \\
&  $\beta_{00} = 0$  & \\
\hline
2 &  $\beta_{22} = (\sim 1, -1, -1, \sim 1, \sim 1, \sim 1)$  & \\
\hline
3 &  $\beta_{ij} = \hat{e} \cdot \hat{\beta}_{ij}$ , & \\
&  $\beta_{00} = 0$  & \\
\hline
\end{tabular}
\end{center}
\end{table}
```

To reference the table in the text, code as follows:

```
Table~\ref{tab1} lists the closed-form...
```

to produce:

Table II lists the closed-form . . .

F. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered and bulleted. There have been numerous options added to IEEEtran to enhance the creation of

Table II. A Simple Table Example.

Order of filter	Arbitrary coefficients e_m	coefficients b_{ij}
1	$b_{ij} = \hat{e} \cdot \beta_{ij},$	$b_{00} = 0$
2	$\beta_{22} = (1, -1, -1, 1, 1, 1)$	
3	$b_{ij} = \hat{e} \cdot \beta_{ij},$	$b_{00} = 0,$

lists. If your lists are more complex than those shown below, please refer to the “IEEEtran_HOWTO.pdf” for additional options.

A plain unnumbered list

bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex

coded as:

```
\begin{list}{}{}{}
\item{bare\_jrnl.tex}
\item{bare\_conf.tex}
\item{bare\_jrnl\_compsoc.tex}
\item{bare\_conf\_compsoc.tex}
\item{bare\_jrnl\_comsoc.tex}
\end{list}
```

A simple numbered list

- 1) bare_jrnl.tex
- 2) bare_conf.tex
- 3) bare_jrnl_compsoc.tex
- 4) bare_conf_compsoc.tex
- 5) bare_jrnl_comsoc.tex

coded as:

```
\begin{enumerate}
\item{bare\_jrnl.tex}
\item{bare\_conf.tex}
\item{bare\_jrnl\_compsoc.tex}
\item{bare\_conf\_compsoc.tex}
\item{bare\_jrnl\_comsoc.tex}
\end{enumerate}
```

A simple bulleted list

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

coded as:

```
\begin{itemize}
\item{bare\_jrnl.tex}
\item{bare\_conf.tex}
\item{bare\_jrnl\_compsoc.tex}
\item{bare\_conf\_compsoc.tex}
\item{bare\_jrnl\_comsoc.tex}
\end{itemize}
```

G. Other Elements

For other less common elements such as Algorithms, Theorems and Proofs, and Floating Structures such as page-wide tables, figures or equations, please refer to the “IEEEtran_HOWTO.pdf” section on “Double Column Floats.”

VI. HOW TO CREATE COMMON BACK MATTER ELEMENTS

The following sections demonstrate common back matter elements such as Acknowledgments, Bibliographies, Appendices and Author Biographies.

A. Acknowledgments

This should be a simple paragraph before the bibliography to thank those individuals and institutions who have supported your work on this article.

```
\section{Acknowledgments}
\noindent Text describing those who
supported your paper.
```

B. Bibliographies

References Simplified: A simple way of composing references is to use the `\bibitem` macro to define the beginning of a reference as in the following examples:

[6] H. Sira-Ramirez. “On the sliding mode control of nonlinear systems,” *Systems & Control Letters*, vol. 19, pp. 303–312, 1992.

coded as:

```
\bibitem{Sira3}
H. Sira-Ramirez. ``On the sliding mode
control of nonlinear systems,’’
\textit{Systems & Control Letters},
vol. 19, pp. 303--312, 1992.
```

[7] A. Levant. “Exact differentiation of signals with unbounded higher derivatives,” in *Proceedings of the 45th IEEE Conference on Decision and Control*, San Diego, California, USA, pp. 5585–5590, 2006.

coded as:

```
\bibitem{Levant}
A. Levant. ``Exact differentiation of
signals with unbounded higher
derivatives,’’ in \textit{Proceedings
of the 45th IEEE Conference on
Decision and Control}, San Diego,
California, USA, pp. 5585--5590, 2006.
```

[8] M. Fliess, C. Join, and H. Sira-Ramirez. “Non-linear estimation is easy,” *International Journal of Modelling, Identification and Control*, vol. 4, no. 1, pp. 12–27, 2008.

coded as:

```
\bibitem{Cedric}
M. Fliess, C. Join, and H. Sira-Ramirez.
``Non-linear estimation is easy,’’
\textit{International Journal of Modelling,
```

Identification and Control}}, vol. 4, no. 1, pp. 12–27, 2008.

[9] R. Ortega, A. Astolfi, G. Bastin, and H. Rodriguez. “Stabilization of food-chain systems using a port-controlled Hamiltonian description,” in *Proceedings of the American Control Conference*, Chicago, Illinois, USA, pp. 2245–2249, 2000.

coded as:

```
\bibitem{Ortega}
R. Ortega, A. Astolfi, G. Bastin, and H.
Rodriguez. ``Stabilization of food-chain
systems using a port-controlled Hamiltonian
description,’’ in \textit{Proceedings of the
American Control Conference}, Chicago,
Illinois, USA, pp. 2245--2249, 2000.
```

C. Accented Characters in References

When using accented characters in references, please use the standard LaTeX coding for accents. **Do not use math coding for character accents.** For example:

```
\'e, \"o, \"a, \"e
```

will produce: é, ö, à, ë

D. Use of BibTeX

If you wish to use BibTeX, please see the documentation that accompanies the IEEEtran Bibliography package.

E. Biographies and Author Photos

Authors may have options to include their photo or not. Photos should be a bit-map graphic (.tif or .jpg) and sized to fit in the space allowed. Please see the coding samples below:

```
\begin{IEEEbiographynophoto}{Jane Doe}
Biography text here without a photo.
\end{IEEEbiographynophoto}
```

or a biography with a photo

```
\begin{IEEEbiography}[{\includegraphics
[width=1in,height=1.25in,clip,
keepaspectratio]{fig1.png}}]
{IEEE Publications Technology Team}
In this paragraph you can place
your educational, professional background
and research and other interests.
\end{IEEEbiography}
```

Please see the end of this document to see the output of these coding examples.

VII. MATHEMATICAL TYPOGRAPHY AND WHY IT MATTERS

Typographical conventions for mathematical formulas have been developed to **provide uniformity and clarity of presentation across mathematical texts**. This enables the readers of those texts to both understand the author’s ideas and to

grasp new concepts quickly. While software such as L^AT_EX and MathType[®] can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article.

As such, IEEE has assembled a set of examples of good and bad mathematical typesetting. You will see how various issues are dealt with. The following publications have been referenced in preparing this material:

Mathematics into Type, published by the American Mathematical Society

The Printing of Mathematics, published by Oxford University Press

The L^AT_EX Companion, by F. Mittelbach and M. Goossens

More Math into LaTeX, by G. Grätzer

AMS-StyleGuide-online.pdf, published by the American Mathematical Society

Further examples can be seen at <http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Math-Typesetting-Guide.pdf>

A. Display Equations

A simple display equation example shown below uses the “equation” environment. To number the equations, use the \label macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^n 2iQ. \quad (22)$$

is coded as follows:

```
\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^n 2{i} Q.
\end{equation}
```

To reference this equation in the text use the \ref macro. Please see (22)

is coded as follows:

```
Please see (\ref{deqn_ex1})
```

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.

Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for sub-equations. This should be consistent throughout the article.

C. Multi-line equations and alignment

Here we show several examples of multi-line equations and proper alignments.

A single equation that must break over multiple lines due to length with no specific alignment.

The first line of this example

The second line of this example

The third line of this example (23)

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$a = c + d \quad (24)$$

$$b = e + f \quad (25)$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The align environment can align on multiple points as shown in the following example:

$$x = y \quad X = Y \quad a = bc \quad (26)$$

$$x' = y' \quad X' = Y' \quad a' = bz \quad (27)$$

is coded as:

```
\begin{align}
x &= y & X &= Y & a &= bc \\
x' &= y' & X' &= Y' & a' &= bz
\end{align}
```

D. Subnumbering

The amsmath package provides a subequations environment to facilitate subnumbering. An example:

$$f = g \quad (28a)$$

$$f' = g' \quad (28b)$$

$$\mathcal{L}f = \mathcal{L}g \quad (28c)$$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f &= g \label{eq:2A} \\
f' &= g' \label{eq:2B} \\
\mathcal{L}f &= \mathcal{L}g \label{eq:2C}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.

A simple matrix:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (29)$$

is coded as:

```
\begin{equation}
\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (30)$$

is coded as:

```
\begin{equation}
\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (31)$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \quad (32)$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad (33)$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a & b \\ c & d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix} \quad (34)$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The `array` environment allows you some options for matrix-like equations. You will have to manually key the fences, but you'll have options for alignment of the columns and for setting horizontal and vertical rules. The argument to `array` controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (35)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (36)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|c|c|c} a+b+c & uv & x-y & 27 \\ \hline a+b & u+v & z & 134 \end{array} \right) \quad (37)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|c|c|c}
a+b+c & uv & x-y & 27\\
\hline
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

Note the argument now has the pipe “|” included to indicate the placement of the vertical rules.

G. Cases Structures

Many times we find cases coded using the wrong environment, i.e., `array`. Using the `cases` environment will save keystrokes (from not having to type the `\left\lbracket`) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
\{z_m(t)\} =
\begin{cases}
```

```
1, & \text{if} \} \{ \beta_m(t), \\
\{0, \} & \text{otherwise.} \} \\
\end{cases}
\end{equation*}
```

Note that the “&” is used to mark the tabular alignment. This is important to get proper column alignment. Do not use `\quad` or other fixed spaces to try and align the columns. Also, note the use of the `\text` macro for text elements such as “if” and “otherwise”.

H. Function Formatting in Equations

In many cases there is an easy way to properly format most common functions. Use of the `\` in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the `\text` macro.

$$d_R^{KM} = \arg \min_{d_1^{KM}} \{d_1^{KM}, \dots, d_6^{KM}\}.$$

is coded as follows:

```
\begin{equation*}
d_R^{KM} = \underset{\{\text{arg min}\}}{\{d_1^{KM}, \dots, d_6^{KM}\}}.
\end{equation*}
```

I. Text Acronyms inside equations

This example shows where the acronym “MSE” is coded using `\text{}` to match how it appears in the text.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

```
\begin{equation*}
\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2
\end{equation*}
```

J. Obsolete Coding

Avoid the use of outdated environments, such as `eqnarray` and `$$` math delimiters, for display equations. The `$$` display math delimiters are left over from PlainTeX and should not be used in L^AT_EX, ever. Poor vertical spacing will result.

K. Use Appropriate Delimiters for Display Equations

Some improper mathematical coding advice has been given in various YouTube™ videos on how to write scholarly articles, so please follow these good examples:

For **single-line unnumbered display equations**, please use the following delimiters:

```
\[ . . . \] or
\begin{equation*} . . . \end{equation*}
```


Note that the `*` in the environment name turns off equation numbering.

For **multiline unnumbered display equations** that have alignment requirements, please use the following delimiters:

```
\begin{align*} . . . \end{align*}
```

For **single-line numbered display equations**, please use the following delimiters:

```
\begin{equation} . . . \end{equation}
```

For **multiline numbered display equations**, please use the following delimiters:

```
\begin{align} . . . \end{align}
```

VIII. L^AT_EX PACKAGE SUGGESTIONS

Immediately after your documenttype declaration at the top of your L^AT_EX file is the place where you should declare any packages that are being used. The following packages were used in the production of this document.

```
\usepackage{amsmath,amsfonts}
\usepackage{algorithmic}
\usepackage{array}
\usepackage[caption=false,font=normalsize,
  labelfont=sf,textfont=sf]{subfig}
\usepackage{textcomp}
\usepackage{stfloats}
\usepackage{url}
\usepackage{verbatim}
\usepackage{graphicx}
\usepackage{balance}
```

IX. ADDITIONAL ADVICE

Please use “soft” (e.g., `\eqref{Eq}`) or (`\ref{Eq}`) cross references instead of “hard” references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please note that the `{subequations}` environment in L^AT_EX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you’ve discovered a new method of counting.

BIB_T_EX does not work by magic. It doesn’t get the bibliographic data from thin air but from .bib files. If you use BIB_T_EX to produce a bibliography you must send the .bib files.

L^AT_EX can’t read your mind. If you assign the same label to a subsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

L^AT_EX does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it’s supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Please do not use `\nonumber` or `\notag` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won’t be any anyway) and it might stop a wanted equation number in the surrounding equation.

X. A FINAL CHECKLIST

- 1) Make sure that your equations are numbered sequentially and there are no equation numbers missing or duplicated. Avoid hyphens and periods in your equation numbering. Stay with IEEE style, i.e., (1), (2), (3) or for sub-equations (1a), (1b). For equations in the appendix (A1), (A2), etc..
- 2) Are your equations properly formatted? Text, functions, alignment points in cases and arrays, etc.
- 3) Make sure all graphics are included.
- 4) Make sure your references are included either in your main L^AT_EX file or a separate .bib file if calling the external file.

REFERENCES

- [1] *Mathematics into Type*, American Mathematical Society. Online available:
- [2] T.W. Chaundy, P.R. Barrett and C. Batey, *The Printing of Mathematics*, Oxford University Press. London, 1954.
- [3] *The L^AT_EX Companion*, by F. Mittelbach and M. Goossens
- [4] *More Math into L^AT_EX*, by G. Grätzer
- [5] *AMS-StyleGuide-online.pdf*, published by the American Mathematical Society
- [6] H. Sira-Ramirez. “On the sliding mode control of nonlinear systems,” *Systems & Control Letters*, vol. 19, pp. 303–312, 1992.
- [7] A. Levant. “Exact differentiation of signals with unbounded higher derivatives,” in *Proceedings of the 45th IEEE Conference on Decision and Control*, San Diego, California, USA, pp. 5585–5590, 2006.
- [8] M. Fliess, C. Join, and H. Sira-Ramirez. “Non-linear estimation is easy,” *International Journal of Modelling, Identification and Control*, vol. 4, no. 1, pp. 12–27, 2008.
- [9] R. Ortega, A. Astolfi, G. Bastin, and H. Rodriguez. “Stabilization of food-chain systems using a port-controlled Hamiltonian description,” in *Proceedings of the American Control Conference*, Chicago, Illinois, USA, pp. 2245–2249, 2000.

Jane Doe Biography text here without a photo.

PLACE
PHOTO
HERE

IEEE Publications Technology Team In this paragraph you can place your educational, professional background and research and other interests.